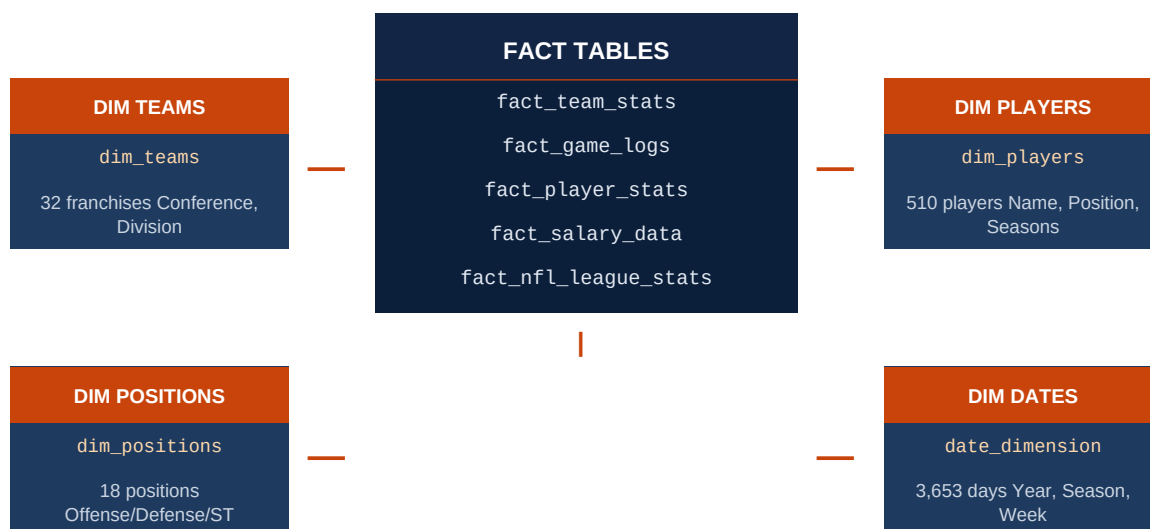


Following the completion of data collection, validation, and dimensional modeling, Phase 3 focused on transforming the analytical datasets into a fully functional relational database. Rather than analyzing individual CSV files directly, the validated data was imported into a SQLite database where SQL was used to verify data integrity, perform exploratory analysis, answer business questions, and create reusable analytical views for Power BI. The objective was to replicate a professional business intelligence workflow by separating data storage, analysis, and reporting into distinct layers.

Step 1 — Database Architecture

The first step was designing the relational database structure for Bearlytics. A SQLite database named `bearlytics.db` was created to serve as the central repository for all validated datasets, structured using the star schema established in Phase 2.



Primary keys and composite keys were defined to ensure data uniqueness while maintaining consistent relationships across the analytical model. This architecture mirrors the structure commonly used in enterprise business intelligence environments.

Step 2 — Database Import Pipeline

An idempotent Python import script was developed to automate loading the validated CSVs into SQLite. Rather than manually importing files, the pipeline can rebuild the entire database from scratch using a single command — eliminating the risk of stale or partially updated data.



Step 3 — Loading the Analytical Model

Nine validated datasets from Phase 2 were imported into SQLite. The completed database serves as the analytical foundation for all future reporting.

Table	Purpose
dim_teams	Team lookup
dim_players	Player dimension
dim_positions	Position lookup
date_dimension	Calendar

Table	Purpose
fact_team_stats	Season stats
fact_game_logs	Game results
fact_player_stats	Player production
fact_salary_data	Contracts
fact_nfl_league_stats	League data

Final database row counts:

		16
	fact_game_logs	5
		19
dim_teams	32	fact_player_stats
dim_players	510	6
dim_positions	18	1,
	3,65	15
date_dimension	3	fact_salary_data
fact_team_stats	10	5
		fact_nfl_league_stat
		s
		32
		0

Step 4 — Database Validation

Once the data was imported, a comprehensive SQL validation process verified the integrity of the database. Because the database originated from validated CSV files, every test passed successfully.

✓ Row count verification	Confirmed every table matched expected Phase 2 record counts
✓ Season range validation	Verified all records fall within 2016–2025
✓ NULL value checks	Confirmed no unexpected missing values in key fields
✓ Duplicate key detection	Confirmed primary key uniqueness across all nine tables
✓ Relationship integrity	Validated foreign key references between fact and dimension tables
✓ Business rule validation	Verified logical constraints such as wins + losses ≤ 17
✓ Dataset profiling	Confirmed distribution and range of key metrics

Step 5 — Exploratory Data Analysis

With the database verified, SQL was used to explore the data before building executive dashboards. Exploratory queries focused on identifying trends, patterns, and relationships throughout the ten-year dataset across four analytical domains.

Team Performance → Highest and lowest scoring seasons → Point differential trends → Offensive and defensive production	Game Analysis → Home vs. away performance splits → Weekly scoring trends → Largest victories and defeats
Player Analysis → Top offensive producers by season → Passing, rushing, and receiving leaders → Career production aggregates	Salary Overview → Contract value distributions → Average contract by position → Position-level salary benchmarks

Step 6 — Business Intelligence Queries

After understanding the data, SQL was used to answer business-focused questions that directly support executive decision making. Fifteen analytical queries were developed across five categories, transforming raw statistics into actionable business insights rather than simply retrieving records.

Executive Performance	Team Performance	Player Performance
→ Best season	→ Offensive improvement	→ Top individual seasons
→ Worst season	→ Defensive improvement	→ Career production totals
→ Season performance tiers	→ Year-over-year trends	→ Offensive contribution
Roster Value	League Comparison	
→ Salary investment	→ Bears vs. NFL average	
→ Contract efficiency	→ Division performance	
→ Cost per team success	→ League ranking	

Step 7 — Reusable SQL Views

Seven production-ready SQL views were created to simplify Power BI reporting and eliminate repeated complex query logic. These views act as reusable analytical datasets that Power BI can query directly.

View Name	Purpose
vw_season_summary	Executive season overview — wins, points, differentials
vw_game_log_detail	Detailed game-level analysis with running record
vw_player_season_stats	Season-level player production by position
vw_player_career_stats	Career totals aggregated across all seasons
vw_salary_efficiency	Salary investment benchmarked against on-field production
vw_bears_vs_league	Bears statistics compared to NFL average by season
vw_home_away_by_season	Home and away performance splits by season

Step 8 — Data Integration Review

During SQL analysis, one remaining integration limitation was identified. The salary dataset stores players using full names, while the player statistics dataset follows the abbreviated naming convention used by nflverse. As a result, not every salary record can be directly joined to player production — primarily affecting defensive players, offensive linemen, specialists, and others not captured in the offensive statistics dataset.

A normalization process was implemented to improve matching where possible, while preserving data accuracy by avoiding ambiguous assignments. Identifying and documenting this limitation demonstrates the importance of understanding source system differences before performing analytical joins — a common challenge in real-world business intelligence work.

Phase 3 Deliverables

At the completion of Phase 3, Bearlytics contains a fully operational relational database with nine tables, seven reusable SQL views, fifteen business queries, and a complete suite of SQL development files.

File	Description	Type
bearlytics.db	Central SQLite analytical database	Database
01_schema.sql	Table definitions, keys, constraints	SQL
02_import.py	Automated CSV import pipeline	Python
03_validate.sql	Database integrity validation	SQL
04_exploratory.sql	Trend and pattern exploration queries	SQL
05_business.sql	15 business intelligence queries	SQL
06_views.sql	7 reusable analytical views	SQL

9	7	15	4	Validation
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Outcome

Phase 3 transforms Bearlytics from a collection of validated datasets into a fully functional analytical database. By implementing a relational schema, automating data imports, validating database integrity, performing exploratory analysis, developing business-focused SQL queries, and creating reusable views, the project now reflects the workflow used by business intelligence and analytics teams in industry.

Rather than relying on spreadsheet calculations or ad hoc analysis, all reporting logic is centralized within SQL. This provides a reliable foundation for the final Power BI dashboards, where the focus shifts from preparing data to communicating insights through interactive visualizations.

Project Roadmap



Green = complete. Orange = current phase. Remaining phases cover Power BI DAX modeling, executive dashboard development, and GitHub portfolio presentation.